

Full code:

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# =====
# 1. GLOBAL DATA
# =====

global_data = {
    "attribute_1": "total_cost",          # always attribute 1
    "attribute_2": "material_continuity", # always attribute 2
    "attribute_3": "total_harness_mass_kg", # lightweighting driver for EV range
    "attribute_4": "electrical_performance_score", # current capacity, voltage drop, signal integ
    "attribute_5": "reliability_durability_score", # vibration, thermal cycling, oxidation, 15yr life

    "direction_attribute_1": "negative", # lower cost is better
    "direction_attribute_2": "positive", # higher material continuity is better
    "direction_attribute_3": "negative", # lower harness mass is better (EV range benefit)
    "direction_attribute_4": "positive", # higher electrical performance is better
    "direction_attribute_5": "positive", # higher reliability is better

    "attribute_1_min": 400, # below cheapest computed product cost (India Al: ~$490)
    "attribute_1_max": 1200, # above most expensive computed product cost (US Cu: ~$1,076)
    "attribute_2_min": 0, # complete material switch (Al replacing Cu)
    "attribute_2_max": 1, # full material continuity (Cu staying Cu)
    "attribute_3_min": 25, # theoretical minimum full-vehicle EV harness (kg)
    "attribute_3_max": 50, # theoretical maximum full-vehicle EV harness (kg)
    "attribute_4_min": 0,
    "attribute_4_max": 10,
    "attribute_5_min": 0,
    "attribute_5_max": 10,

    "copper_price_per_ton": 9500, # fixed global Cu price
    "aluminum_price_per_ton": 2500, # fixed global Al price
    "copper_price_per_kg": 9.500, # fixed global Cu price
    "aluminum_price_per_kg": 2.500, # fixed global Al price
}

# =====
# 2. REGIONAL DATA
# =====

regional_data = [
    {
        "region": "United States",
        "copper_product_market_share": 0.85, # Cu harness dominant; some OEM Al pilots
        "aluminum_product_market_share": 0.15,
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"weight_attribute_1": 0.25, # cost important but not dominant; premium + mass-mar
"weight_attribute_2": 0.15, # moderate conservatism on material switching
"weight_attribute_3": 0.20, # range anxiety culture keeps lightweighting meaningful
"weight_attribute_4": 0.20, # NHTSA and OEM quality standards maintain perf
"weight_attribute_5": 0.20, # long service life expectations (15+ years)
},
{
  "region": "China",
  "copper_product_market_share": 0.75, # growing AI adoption (BYD, NI
  "aluminum_product_market_share": 0.25,
  "weight_attribute_1": 0.35, # most price-competitive EV market globally
  "weight_attribute_2": 0.15, # lower lock-in; domestic Tier-1s actively dev
  "weight_attribute_3": 0.20, # range competitiveness important in dense EV market
  "weight_attribute_4": 0.15, # performance important but subordinate to cost
  "weight_attribute_5": 0.15, # reliability secondary to cost at current market maturity
},
{
  "region": "India",
  "copper_product_market_share": 0.92, # very conservative; Cu deeply e
  "aluminum_product_market_share": 0.08,
  "weight_attribute_1": 0.45, # economy EV market; ₹/km is the governing metric
  "weight_attribute_2": 0.20, # highest conservatism on switching; incumbent Cu lock-in
  "weight_attribute_3": 0.10, # simpler shorter-range vehicles; weight penalty less critical
  "weight_attribute_4": 0.13, # electrical performance still needed but simp
  "weight_attribute_5": 0.12, # reliability important but cost pressures dominate trade-offs
},
{
  "region": "Europe",
  "copper_product_market_share": 0.80, # lightweighting pressure from EU
  "aluminum_product_market_share": 0.20,
  "weight_attribute_1": 0.20, # cost matters but quality trade-offs more acceptable than Asia
  "weight_attribute_2": 0.15, # REACH/sustainability regulations add nua
  "weight_attribute_3": 0.25, # EU fleet CO2 regulations make we
  "weight_attribute_4": 0.20, # premium OEM culture (BMW, Mercedes, VW)
  "weight_attribute_5": 0.20, # long expected vehicle lifetimes; high reliability standard
},
]

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# =====
# 3. PRODUCT DATA
# =====
product_data = [

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— UNITED STATES —————

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{
  "region": "United States",
  "dominant material": "cu",
  "product": "EV Wire Harness",
  "copper_kg": 18,      # primary conductor material throughout full harness
  "aluminum_kg": 2,     # minor Al in mounting clips, brackets
  "total_mass_kg": 40,  # ~20 kg metallic + ~20 kg insulation/connectors/sleeving
  "non_material_cost_per_unit": 900, # high US labor + overhead; assembly s
  "attribute_2_value": 1, # Cu is the global incumbent harness material — full continuity
  "attribute_3_value": 40, # total harness mass (kg); directly impacts EV range
  "attribute_4_value": 9,  # Cu sets the electrical performance benchmark (0–10)
  "attribute_5_value": 9,  # proven 15-yr durability; excellent vibration/oxidation resistance
},
{
  "region": "United States",
  "dominant material": "al",
  "product": "EV Wire Harness",
  "copper_kg": 4,      # fine signal wires (<0.5mm²) + connector pins retained in Cu
  "aluminum_kg": 7,    # Al replaces medium/large gauge power
  "total_mass_kg": 31, # ~22% lighter than Cu variant;
  "non_material_cost_per_unit": 960, # +$60 premium vs Cu: bi-m
  "attribute_2_value": 0, # switching from Cu incumbent — zero material continuity
  "attribute_3_value": 31, # lighter harness; ~9 kg saved vs Cu variant
  "attribute_4_value": 7,  # meets specs with derating; some signal inte
  "attribute_5_value": 7,  # oxidation at terminations, cold flow un
},

{
  "region": "China",
  "dominant material": "cu",
  "product": "EV Wire Harness",
  "copper_kg": 18,
  "aluminum_kg": 2,
  "total_mass_kg": 40,
  "non_material_cost_per_unit": 500, # lower labor cost; China is
  "attribute_2_value": 1,
  "attribute_3_value": 40,
  "attribute_4_value": 9,
  "attribute_5_value": 9,
},
{
  "region": "China",
  "dominant material": "al",

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    "product": "EV Wire Harness",
    "copper_kg": 4,
    "aluminum_kg": 7,
    "total_mass_kg": 31,
    "non_material_cost_per_unit": 555, # +$55 vs Cu China; lower absolute p
    "attribute_2_value": 0,
    "attribute_3_value": 31,
    "attribute_4_value": 7,
    "attribute_5_value": 7,
  },

  {
    "region": "India",
    "dominant material": "cu",
    "product": "EV Wire Harness",
    "copper_kg": 18,
    "aluminum_kg": 2,
    "total_mass_kg": 40,
    "non_material_cost_per_unit": 380, # very low labor cost; Tata, M
    "attribute_2_value": 1,
    "attribute_3_value": 40,
    "attribute_4_value": 9,
    "attribute_5_value": 9,
  },

  {
    "region": "India",
    "dominant material": "al",
    "product": "EV Wire Harness",
    "copper_kg": 4,
    "aluminum_kg": 7,
    "total_mass_kg": 31,
    "non_material_cost_per_unit": 435, # +$55 vs Cu India; termination system cost premium
    "attribute_2_value": 0,
    "attribute_3_value": 31,
    "attribute_4_value": 7,
    "attribute_5_value": 7,
  },

  {
    "region": "Europe",
    "dominant material": "cu",
    "product": "EV Wire Harness",
    "copper_kg": 18,

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"aluminum_kg": 2,
"total_mass_kg": 40,
"non_material_cost_per_unit": 850, # high EU labor + premium OEM quality overhead
"attribute_2_value": 1,
"attribute_3_value": 40,
"attribute_4_value": 9,
"attribute_5_value": 9,
},
{
  "region": "Europe",
  "dominant material": "al",
  "product": "EV Wire Harness",
  "copper_kg": 4,
  "aluminum_kg": 7,
  "total_mass_kg": 31,
  "non_material_cost_per_unit": 910, # +$60 vs Cu Europe; premium c
  "attribute_2_value": 0,
  "attribute_3_value": 31,
  "attribute_4_value": 7,
  "attribute_5_value": 7,
},
]
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